Hadrian Lazic

(Rust Low-Level Systems Optimization)

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PROFESSIONAL SUMMARY

Performance-focused Systems Programmer and Technical Evaluator with 5+ years of deep Rust engineering experience and over 23,270 open-source crate downloads on Crates.io. Published machine learning researcher with expertise in microarchitecture optimization, unsafe Rust invariants, L1/L2 cache residency, and hardware-software co-design. Adept at technical code auditing, edge-case generation, API design, and evaluating complex LLM code execution for correctness and memory safety.

TECHNICAL SKILLS

- Rust (5 Years); unsafe, Raw Pointers, SIMD, Inline ASM, RISC-V Assembly, SQLite and Python.
- Systems Architecture and Performance: Mechanical Sympathy, Memory Hierarchy (L1/L2 Cache Residency, 64-Byte Alignment), CPU Pipeline Optimization (Hardware Prefetching, Branchless Programming), Instruction-Level Parallelism (ILP), Arena Memory Allocation.
- AI Evaluation and Technical Writing: Code Auditing, Edge-Case Prompt Engineering, Technical Documentation, Static/Dynamic Analysis, ML Research Publishing.
- Tooling and Environments: Cargo, Linux (Fedora, FreeBSD, Debian), Git, Neovim/LSP, LLVM, GDB/LLDB, perf, Valgrind.

PROJECTS

Laststraw & Laststraw-core — *Solo Author & Maintainer (17,800+ Downloads)*

- **Native Desktop Framework:** Engineered a lightweight, high-performance GUI framework in Rust to replace resource-heavy Electron and WASM browser runtime engines.
- **API Design & Adoption:** Designed ergonomic, zero-cost public APIs and authored detailed technical documentation, driving 17,800+ downloads on Crates.io.

TerimalRtdm — *Solo Author & Maintainer (3,000+ Downloads)*

- **Low-Latency TUI Engine:** Built an asynchronous Terminal UI library utilizing direct ASCII/ANSI escape code manipulations for zero-overhead rendering.
- **Pipeline Saturation:** Optimized internal event loops to eliminate CPU pipeline stalls and maintain a minimal memory footprint.

BitmaskWindow & Zavod-Lang — *Published Rust Crates & VM Runtimes*

- **Bitmask Generator (`BitmaskWindow`):** Created a zero-overhead, data-oriented bitmask library using compile-time bitwise operations to isolate data windows without dynamic allocations.
- **Opcode Interpreter (`Zavod-Lang`):** Built a high-throughput execution virtual machine and opcode interpreter engineered for L1 cache-friendly instruction fetch-decode-execute cycles.

Machine Learning Research Preprint — *Solo Researcher & Author*

- **AI/ML Technical Publication:** Authored and published independent machine learning research on *Preprints.org*, focusing on model behavior, specialized execution structures, and algorithmic evaluation.

Education

B.S. in Computer Science — *Incoming Fall 2026*

San Diego State University (SDSU) – San Diego, CA

- *Focus:* Computer Architecture, Systems Programming, Data Structures & Algorithms, Hardware-Software Co-Design.